

Problem Statement

We need to build a mobile application that records clinical conversations and delivers Abridge's AI-generated clinical notes to the end users.

The app needs to record and upload audio during a clinical conversation, and when the user marks the end of a conversation, it should generate a clinical note from the recorded audio and display it to the user. Furthermore, the user should also be able to pause and resume recording during a conversation as needed.

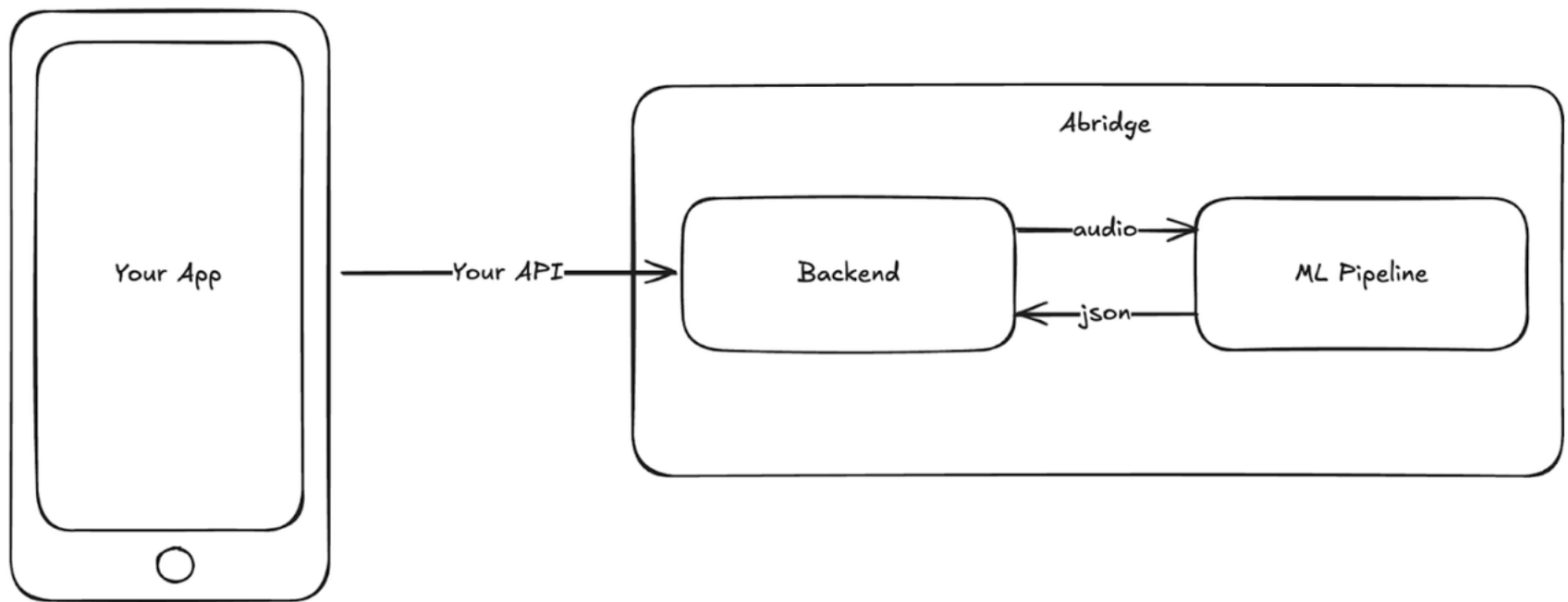
Let's assume streaming audio is not supported due to the limitations of our ML pipeline.

You have been tasked with designing and implementing this feature from scratch.

This is primarily a discussion-based exercise. You don't have to write any code. Pseudocode or written outline of the solution is fine.

Please feel free to discuss and ask about the problem statement, existing infrastructure, and requirements.

You can use this Google Doc, or any tool that you prefer to share your design and ideas with.



Clarifying Questions:

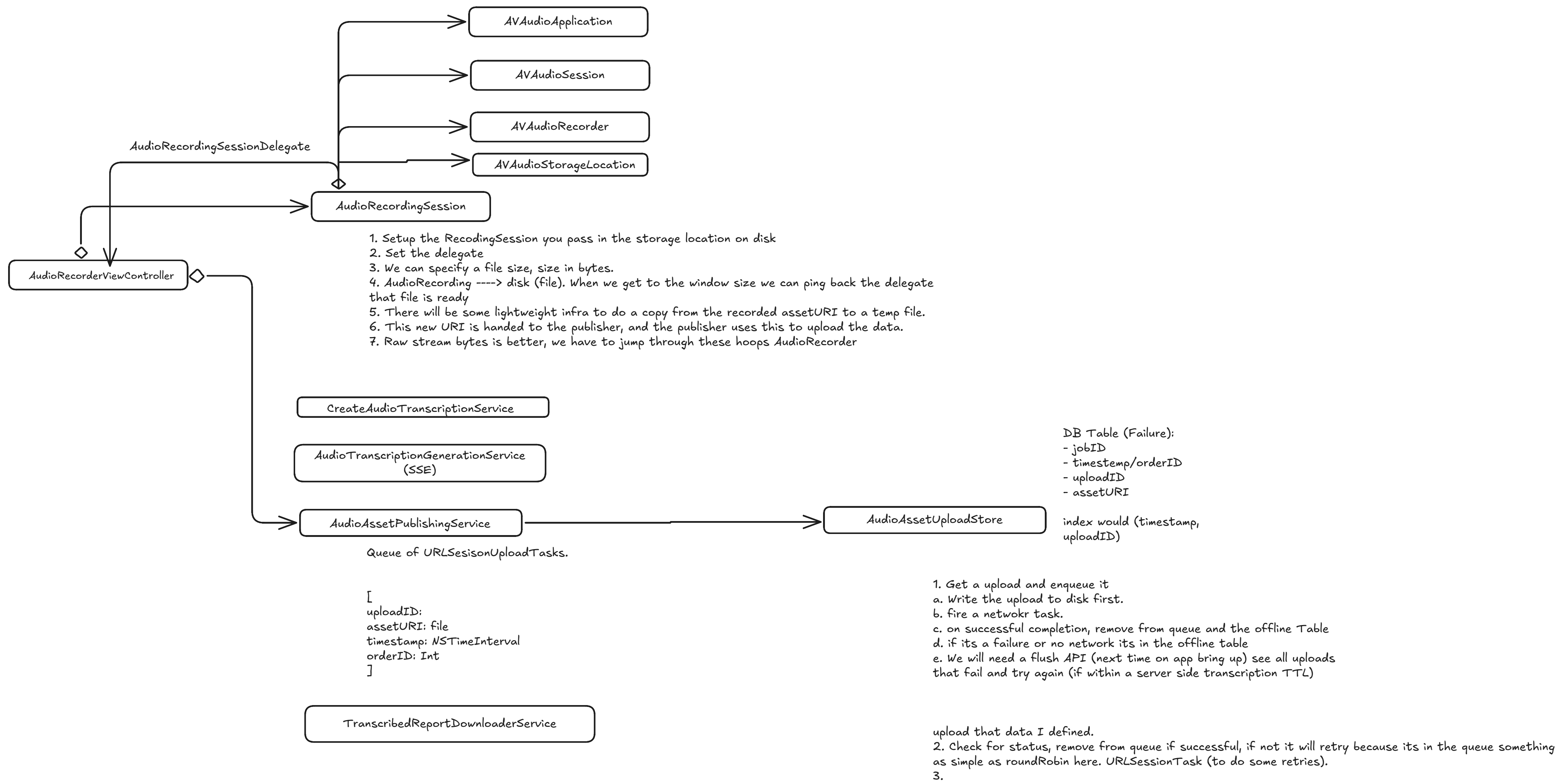
1. Single Recording Screen where you kick off recording and get a analyzed transcript. Then atleast be able to display the recording and transcript as a preview to the user.
2. List view and managing an array of recordings or audio files out of scope, single vanilla screen

Functional Reqs:

1. Single UI with ability to record a conversation, pause, stop the conversation.
2. Ability to convert an audio recording into a speech-recognized transcript.
3. We need not support a real time stream, so we can assume the server side API and ML Model works based off of entire audio asset files.
4. Recording and Batch uploading is important here.

Non-Funcational Reqs:

1. Clinical conversations can take a long time, althought there is no real time stream/tokenization API we dont want to wait for an entire recording.
2. Performance of uploading Audio files and recordings is important, we need to be able to upload quickly and in the background, and get transcriptions.



1. createTranscription endpoint
https://abridge.com/create_transcription POST

```
{
  transcriptionID: "87659"
}
```

2. audioAssetPublishing endpoint
https://abridge.com/transcription_asset_upload POST

```
{
  upload: {bytes of the audio/encoding}
}
Post Body: {
  transcriptionID: "87659"
}
```

3. generateTranscription HTTP POST
https://abridge.com/generate_transcription?transcription_id=87659

```
{
  reportID: "7756"
}
```

4. SSE. POST (Server side endpoint, realtime update from server to client when transcription done)
https://transcriptionReportSession?reportID: 7756

Delivered to client in real-time when report is done

5.

```
{
  reportReady
  [cdnURL: https://report_id] //could be array of URLs to download transcribed report if report is large (chunked)
}
```

6. Download the report from the endpoint and render it.